CELLA: A 28nm Compute-Memory Co-Optimized Real-Time Digital CIM-based Edge LLM Accelerator with 1.78ms-response in Prefill and 31.32 Token/s in Decoding

Zihan Wu[1], Yiqi Wang[1], Zhen He[1], Bohan Yang[2], Yang Wang[1], Shaojun Wei[1], Yang Hu[1], Fengbin Tu[2]*, Shouyi Yin[1]*
[1]Tsinghua University, Beijing; [2]The Hong Kong University of Science and Technology, Hong Kong; *Corresponding Authors

Abstract

This work presents a digital Compute-in-memory (CIM)-based Edge LLM Accelerator CELLA with Compute-Memory co-optimization for *Prefill*-*Decoding* phases: **1)** *Prefill* is compute-bound and CIM works in MAC-mode. A puzzle-shaped input scheduling unit (PISU) reassembles input into an interlocking pattern, reducing 2.82× TTFT. **2)** *Decoding* is memory-bound and CIM works in CAM-mode. **(a)** For *static* parameters, non-uniform quantization multiply-accumulation (NUQ-MAC) is processed in CIM with majority-first index searcher (MFIS), saving 4.89× TPOT. **(b)** For *dynamic* KV cache, only important KV is fetched by hash-based cache filter (HCF), saving 4.13× TPOT. It reaches 1.78ms-response and 31.32token/s, speeding up 2.53~9.74× than state-of-the-art accelerators and edge GPU.

Introduction

Transformer-based Large Language Model (LLM) stands as a revolutionary breakthrough of AI. Unlike prior transformers, LLM has 2 phases: *Prefill* and *Decoding*. *Prefill* generates the first token by summarizing human prompt, which is evaluated by Time-To-First-Token (TTFT). *Decoding* auto-regressively outputs tokens, whose time interval is called Time-Per-Output-Token (TPOT). Digital CIM is efficient for transformers [1-3]. Yet, *Prefill* and *Decoding* exhibit different bound properties, raising **challenges** for instant response: **1)** *Prefill* is compute-bound. Due to outliers [5-8], LLM's input needs floating-point (FP) whose alignment incurs extra TTFT in CIM. **2)** *Decoding* is memory-bound since each token generation requires loading all *static* parameters and *dynamic* KV cache. Memory-efficient algorithms can reduce TPOT: **(a)** *Static* parameters with NUQ [9-11] can be transferred with low bitwidth and computed with high precision; **(b)** *Dynamic* KV cache with hash prediction [12] only fetches important KV from off-chip. Both of them are based on search or comparison, but previous CIM can't process these operations due to internal direct accumulation structure.

Proposed CIM-based Edge LLM Accelerator

Fig.3 shows CELLA's architecture, including 16 CIM cores (each has 4 macros) and an HCF. Each macro has 32 arrays, a PISU, and an MFIS. Fig.4 shows the workflow and CIM core's circuit-level work modes. *Static* parameters with INT8 (*Prefill*) and sub-4bit NUQ (*Decoding*) are stored in DRAM. *Prefill*-*Decoding* co-optimization is achieved by following **features**: **1)** In *Prefill*, CIM works in MAC-mode which stores INT8 parameters for higher performance. PISU loads prompt tokens with FP and reassembles aligned bits into interlocking pattern, reducing TTFT caused by FP alignment. **2)** In *Decoding*, CIM works in CAM-mode for less memory access. **(a)** For QKV-Gen. and FFN layers, *static* parameters with sub-4bit NUQ are loaded in CIM and directly computed by index search, saving TPOT by low-bitwidth transmission. MFIS adjusts index search order for fewer mismatches, further improving the NUQ-MAC efficiency. **(b)** For attention layers, all *dynamic* K is kept in CIM as low-dimensional hash projection (HC_K). HCF projects current q to hash and compares it with HC_K to predict attention scores, saving TPOT by only fetching the important KV cache.

Feature1: CIM's MAC-mode for *Prefill*

Fig.5 and Fig.6 show PISU for FP-INT-MAC, reducing TTFT caused by FP alignment. Since LLM's input has channel-wise similarity [13], INT8 weights are pre-distributed to keep input channel descending. FP16 inputs are aligned to E_{INmax} and converted to 2C. All input bits are cut into puzzle-shaped pieces. Sign piece is transferred to corresponding weight by activating W or $\overline{W}$, eliminating explicit feeding. Additional 1 is added to $\overline{W}$ to transform it to -W. For dense section, CIM directly adds all the partial products. Since MSB piece and LSB piece are caused by shifting, the tabs of MSB piece fit perfectly into the slots of LSB piece, forming interlocking section. Thus, partial products are divided into two parts in each cycle, which are first added separately and then shift-accumulated for correct result. For example, CIM computes $\{IN_2[6]\times(W_2+1)+IN_1[9]\times W_1+IN_{14}[h]\times(W_{14}+1)\}<<11+\{...+IN_0[2]\times W_0\}$@cycle 8. PISU saves 2.82× TTFT and 2.73× energy with 4.6% area overhead.

Feature2: CIM's CAM-mode for *Decoding*

A. MFIS for static parameters: Fig.7 details MFIS for NUQ-MAC, saving TPOT by loading parameters with low bitwidth. Parameters are offline grouped and quantized into a sub-4bit (n=1~4 for different groups) index table and 2^n representative data (RD). Taking 2bit NUQ as an example, indices (ID_0~ID_{15}) are vertically stored across CIM banks (Fig.9 top). The last 2 rows are not used thus gated. According to the distributive law, all inputs corresponding to each index should be accumulated separately. As shown in step 0, index 00 is fed to wordlines for search and all inputs are ANDed with their corresponding match signal then added up by the adder tree. Thus, all inputs corresponding to index 00 are accumulated (e.g. IN_1+IN_5). Mismatch consumes more energy than match due to discharge. Many mismatches are redundant since it is already matched at previous step, which can be gated. According to commutative law, MFIS adjusts index search order by majority-first strategy so that the index with more number will be matched in earlier step, minimizing total mismatches. Lastly, accumulated results are multiplied with RD_0~RD_3 to get the final results. MFIS saves 4.89× TPOT and 4.83× energy with 2.9% area overhead.

B. HCF for dynamic KV cache: Fig.8 depicts HCF for hash prediction, saving TPOT by only fetching important KV cache. KV cache expands linearly during *Decoding*. Prior accelerators with pre-defined sparse attention [1-3] can't be used. Instead of loading all KV cache from off-chip, hash-based method [12] predicts attention scores to identify important KV cache. All previous K is kept in CIM as low-dimensional hash projection (HC_K). Taking *m*-th k-vector as an example, k_m is projected to 16-bit hash code (HC_{km}) and stored in the same row of different CIM banks (Fig.9 bottom), while other rows are gated. At runtime, current q_i is multiplied with projection vectors (p_0~p_{15}) to get the hash code (HC_{qi}). HC_{qi} is fed to wordlines for bit-wise comparison with HC_{km}. Match signals are ANDed with 1 then added up by the adder tree to get the hamming distance, which can be translated to the cosine of the angle between q_i and k_m by LUT. Multiplied with corresponding k norm ($||k_m||$), the predicted attention scores are obtained. After comparing with a threshold, important KV cache is identified and then fetched from DRAM to the KV cache buffer (KVCB). HCF saves 4.13× TPOT and 3.22× energy with 5.3% area overhead.

Measurement Results

Fig.10 shows the fabricated 28nm CELLA that works at 0.6-1.0V, 110-500MHz. Off-chip DDR3 is included for end-to-end evaluation. For OPT-1.3B with various *Prefill*/*Decoding* ratios, it reduces 2.51-5.38× TTFT in *Prefill* and 6.13-9.74× TPOT in *Decoding*, with the breakdown analysis on the bottom left. It reaches 1.78ms-response and 31.32 token/s @1.0V, 500MHz. Compared with state-of-the-art accelerators, CELLA features Compute-Memory co-optimization for both *Prefill*-*Decoding* phases, obtaining TTFT/TPOT speedup of 4.28×/5.20× over [3] and 4.81×/4.04× over [4]. Thanks to CIM's CAM-mode that significantly reduces bandwidth requirement, it obtains 2.53× TPOT speedup than the more advanced 16nm edge GPU [14].

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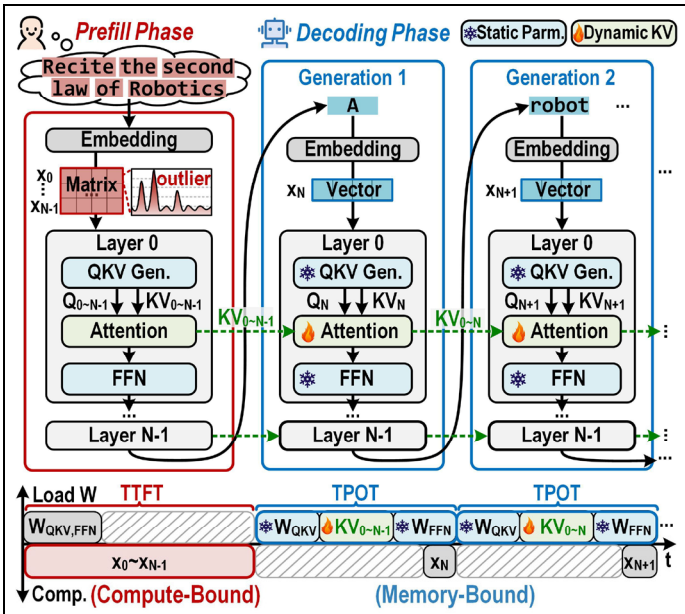


Fig. 1. Prefill and Decoding exhibit different bound properties.

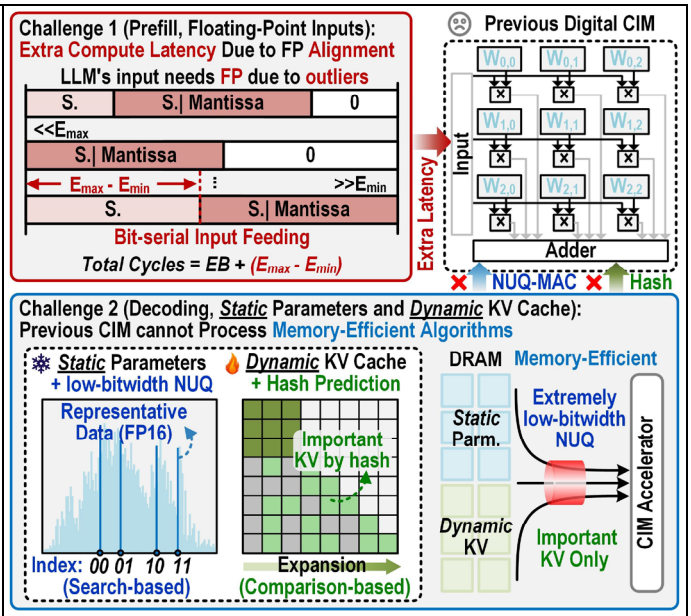


Fig. 2. Prefill and Decoding raise new challenges in digital CIM.

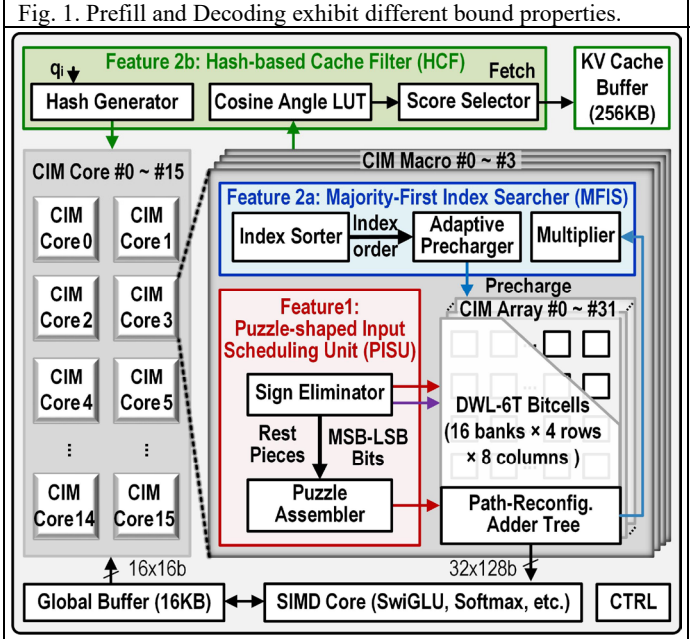


Fig. 3. Overall architecture of CELLA and its key features.

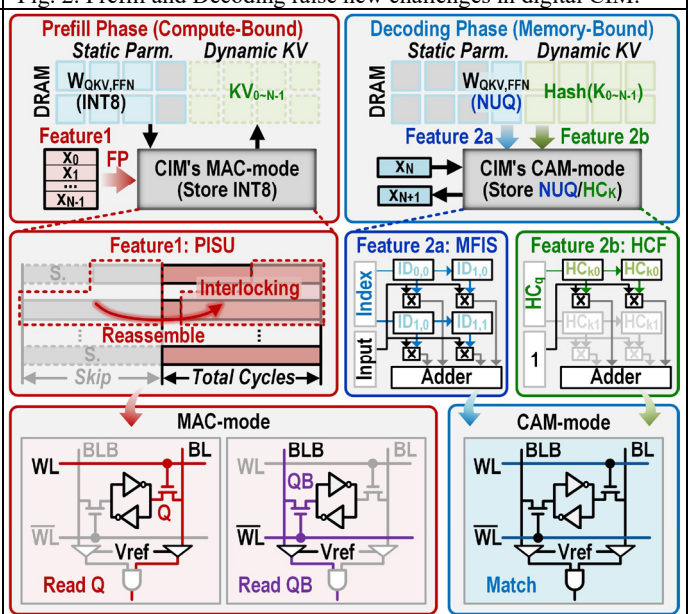


Fig. 4. Prefill/Decoding workflows and CIM's corresponding modes.

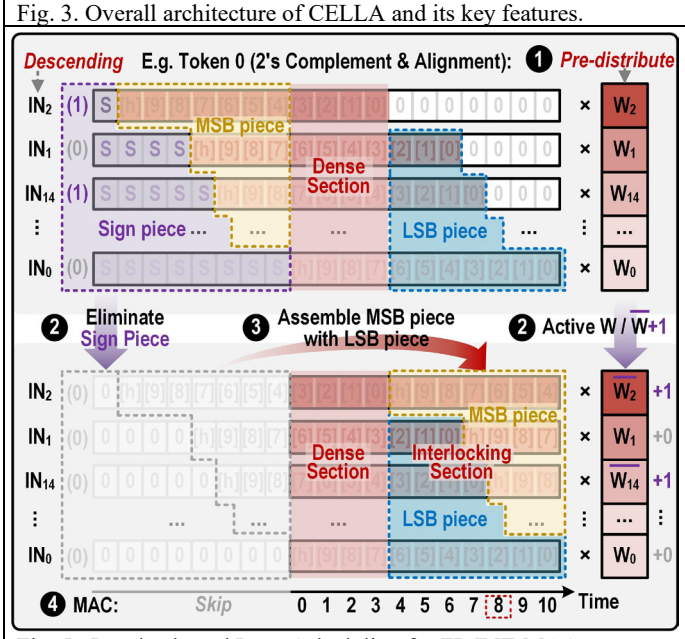


Fig. 5. Puzzle-shaped Input Scheduling for FP-INT-MAC.

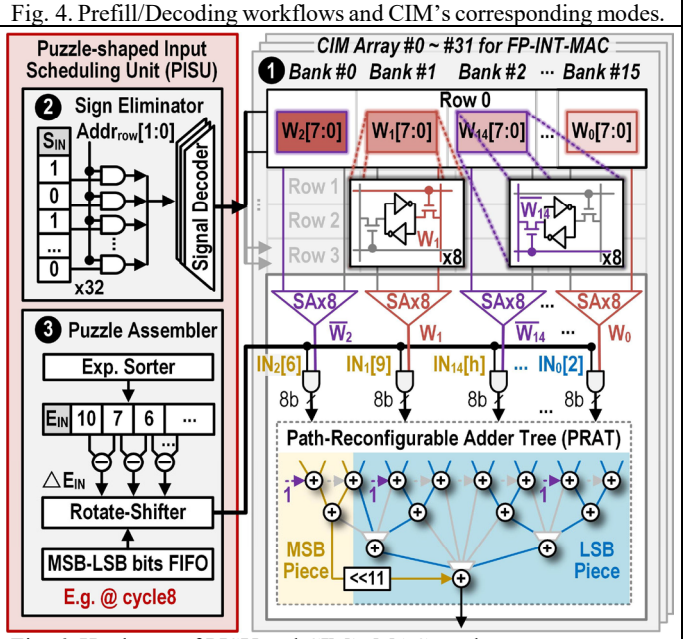


Fig. 6. Hardware of PISU and CIM's MAC-mode.

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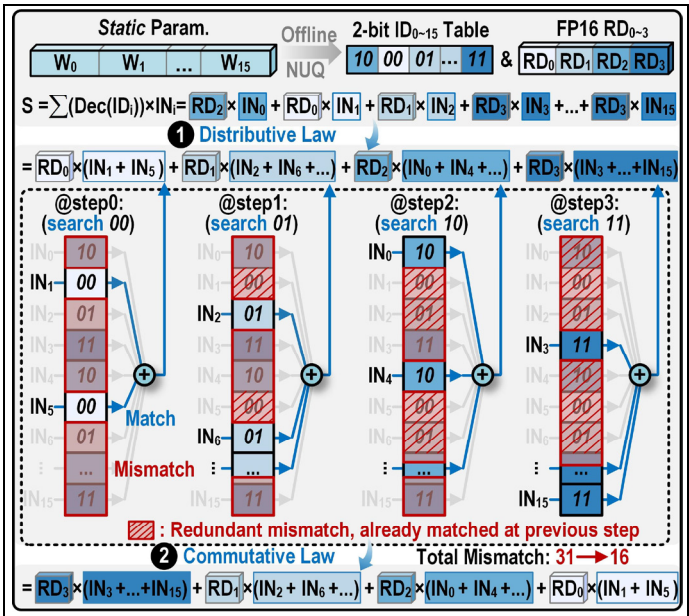


Fig. 7. Majority-first index searching for NUQ-MAC.

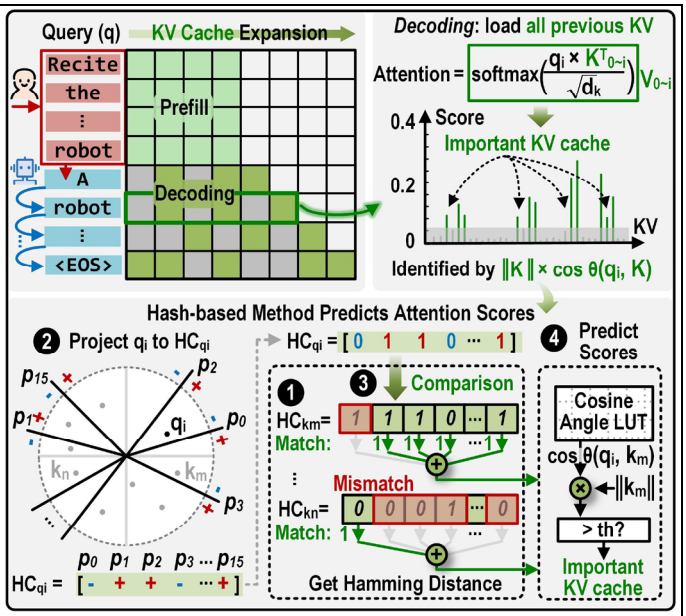


Fig. 8. Hash prediction for important KV cache fetching.

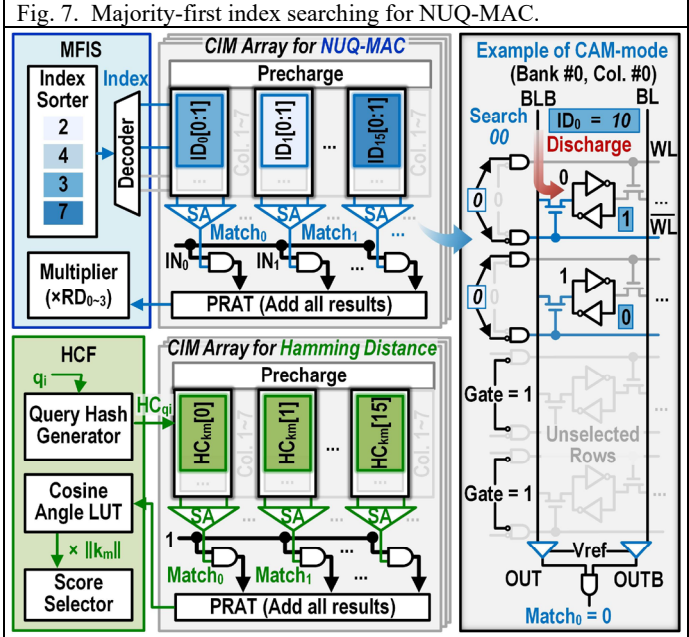
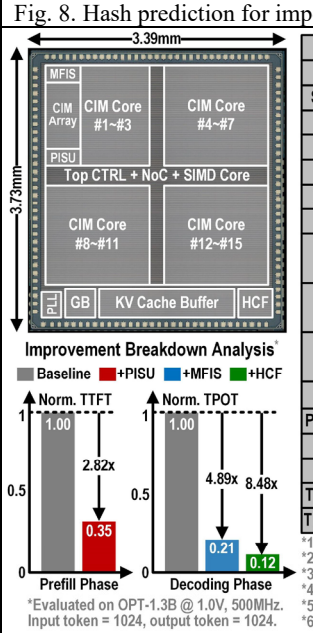


Fig. 9. Hardware of MFIS/HCF and CIM's CAM-mode.



*Evaluated on OPT-1.3B @ 1.0V, 500MHz. Input token = 1024, output token = 1024.

Technology	TSMC 28nm CMOS
Die Area	12.65mm²
Supply Voltage	0.6V-1.0V
Frequency	110MHz-500MHz
Buffer Size	16KB GB + 256KB KVCB
CIM Size	128KB
Data Precision	FP16-INT8/FP16-Sub4-NUQ
Work Mode	MAC-mode/CAM-mode
Power	56.74mW @0.6V, 110MHz
	335.04mW @1.0V, 500MHz
Peak Performance*1	FP-INT-MAC: 2.98TOPS
	NUQ-MAC*3: 0.12-0.96TOPS
Energy Efficiency*2	FP-INT-MAC: 15.6TOPS/W
	NUQ-MAC*3: 0.85-14.1TOPS/W
Model	OPT-1.3B / LLaMA2-7B
Perplexity Loss*4	+0.48 (↓) / +0.79 (↓)
TTFT*1*5	1.78ms-response / 7.20ms-response
1 / TPOT*1	31.32 token/s / 8.51 token/s
TTFT Speedup*1*6	2.51×-5.38× / 3.43×-6.62×
TPOT Speedup*1*6	6.13×-9.74× / 7.96×-10.35×

*1: At the highest performance point, 1.0V, 500MHz.
*2: At the highest energy efficiency point, 0.7V, 245MHz.
*3: Different bitwidths for sub-4bit NUQ (n = 1~4bit).
*4: Compared with FP32. Lower perplexity is more accurate.
*5: "-response" means average response time per input token.
*6: Prefill/Decoding ratios = 1/128, 1, 128 for different tasks.

Fig. 10. Measurement results, die photo and summary table.

TABLE I. Comparison with state-of-the-art Transformer accelerators

	NVIDIA Jetson-TX2	ISSCC'23 16.1 [2]	ISSCC'23 16.2 [3]	ISSCC'24 20.5 [4]	This work
Implementation	Digital Logic	Digital CIM	Digital CIM	Digital Logic	Digital CIM
Phase Optimization	✗	Only Prefill	Only Prefill	Only Prefill	✓ Prefill & Decoding
Work Mode	MAC	MAC	MAC	MAC	✓ MAC/CAM
Precision	FP16/FP32	INT8/INT16	INT8	INT8	✓ FP16-INT8 &Sub4-NUQ
Technology	16nm	28nm	28nm	28nm	28nm
Die Area (mm²)	N/A	14.36	3.93	20.25	12.65
Frequency (MHz)	1300	85-275	20-320	50-200	110-500
Supply Voltage (V)	N/A	0.6-1.0	0.64-1.03	0.7-1.1	0.6-1.0
Power (mW)	15000	29.83-152.75	8.27-250.65	47.5-469.2	56.74-335.04
Peak Performance (TOPS or TFLOPS)	1.33(FP16)	0.89(INT16)	0.83*3(INT16)	0.85*3(INT16)	2.98*1(FP-INT) 0.12-0.96*1*4(NUQ)
Energy Efficiency (TOPS/W or TFLOPS/W)	0.089(FP16)	12.1(INT16)	6.31*3(INT16)	11.9*3(INT16)	15.6*2(FP-INT) 0.85-14.1*2*4(NUQ)
TTFT*5*6 (ms-response↓)	0.37 (0.2×)	4.56 (2.56×)	7.62 (4.28×)	8.56 (4.81×)	1.78*1
1 / TPOT*5*7 (token/s↑)	12.37 (2.53×)	9.73 (3.21×)	6.02 (5.20×)	7.75 (4.04×)	31.32*1

*1: At the highest performance point, 1.0V, 500MHz. *2: At the highest efficiency point, 0.7V, 245MHz.
*3: To satisfy the precision requirement of LLM, maximum reported value of [3-4] is normalized to INT16.
*4: Static parameters with sub-4bit NUQ have different quantized bitwidths (n = 1~4bit) for different groups.
*5: Evaluated on OPT-1.3B. Input token = 1024, output token = 1024. Off-chip memory (DDR3) is included.
*6: "-response" means average response time per input token. *7: Average token generation throughput.

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